

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I Harshavardhan RV- 192472163 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Harshavardhan R V

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c\log(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

2 4

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 4 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

Answers:

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation is an image enhancement technique used to modify the brightness and contrast of an image by changing the intensity values of its pixels. It helps improve image visibility, highlight important details, and prepare images for further processing.

(b) Negative Transformation

Formula:

$$s = 255 - r$$

Given:

$$r = [150 \ 200]$$

Calculation:

- **For $r = 50$:**
- $s = 255 - 50 = 205$

- **For $r = 100$:**
- $s = 255 - 100 = 155$

- **For $r = 150$:**
- $s = 255 - 150 = 105$

- **For $r = 200$:**
- $s = 255 - 200 = 55$

Transformed Values

Original (r)	Transformed (s)
50	205
100	155
150	105
200	55

Answer: [105 55]

2. Log Transformation

(a) Purpose

Log transformation enhances low-intensity (dark) pixels by expanding their values while compressing higher intensity values. This makes details in dark regions more visible.

(b) Calculation

Formula:

$$s = c \log(1 + r)$$

Given:

- $c = 10$
- $r = [100]$

Using natural logarithm ($\ln$):

For $r = 0$ $s = 10\ln(1) = 10(0) = 0$
--

For $r = 10$ $s = 10\ln(11) = 10(2.398) = 23.98$
For $r = 50$ $s = 10\ln(51) = 10(3.932) = 39.32$
For $r = 100$ $s = 10\ln(101) = 10(4.615) = 46.15$

Result

r	s
0	0.00
10	23.98
50	39.32
100	46.15

Answer:

[39.32 46.15]

3. Histogram Processing

(a) Purpose of Histogram Equalization

Histogram equalization improves image contrast by redistributing pixel intensity values over the entire intensity range. This makes dark regions brighter and bright regions clearer, revealing more image details.

(b) Histogram Equalization

Given Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

For a 3-bit image,

$$L = 8$$

Equalization Formula

$$s_k = (L - 1) \times CDF$$

where

$$L - 1 = 7$$

Step 1: Compute Cumulative Distribution

Intensity	Frequency	Cumulative Frequency	CDF
0	2	2	2/16=0.125
1	2	4	4/16=0.25
2	4	8	8/16=0.50
3	8	16	16/16=1.00

Step 2: Equalized Intensities

$$s = 7 \times CDF$$

Intensity	CDF	Equalized Value
0	0.125	0.875≈1
1	0.25	1.75≈2
2	0.50	3.5≈4
3	1.00	7

Final Answer

Original Intensity	Equalized Intensity
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Purpose

Spatial smoothing filters reduce image noise and small variations by replacing each pixel with the average of its neighboring pixels. This produces a smoother image.

(b) Apply 3×3 Averaging Filter

Matrix

10 20 30 20 30 40 30 40 50

Average Filter

Sum of all pixels

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

Answer

The new center pixel value is

30

5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening filters enhance image edges and fine details by emphasizing intensity differences between neighboring pixels. This makes the image appear clearer and more detailed.

(b) Laplacian Filter

Mask

$$\begin{bmatrix} 0 & -1 & 0 & -1 & 0 \\ 0 & -1 & 4 & -1 & 0 \\ 0 & -1 & 0 & -1 & 0 \end{bmatrix}$$

Image

10 10 10 10 20 10 10 10 10

Calculation

$$= (4 \times 20) - (10 + 10 + 10 + 10) = 80 - 40 = 40$$

Answer

The output at the center pixel is

40
